

N 85

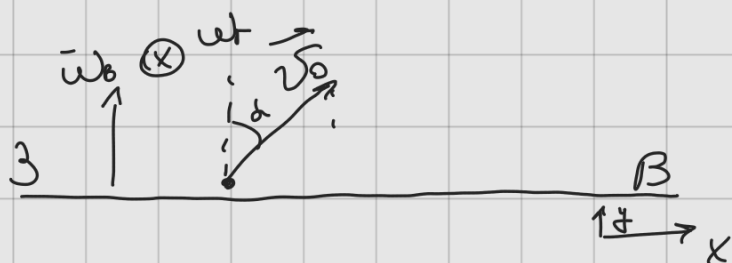
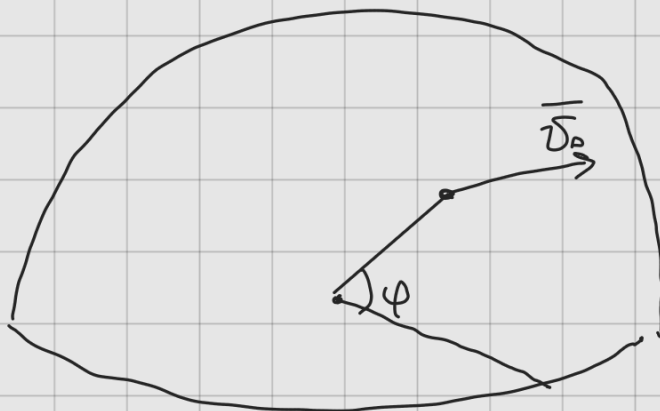
Дано:

$$v_0 = 100 \text{ м/с}$$

$$\varphi = 60^\circ$$

Найти: Δ

Решение:



$$1) \quad \omega r = \omega \cos \varphi ; \quad v_y = v_0 \cos \alpha - gt$$

$$v_x = v_0 \sin \alpha$$

$$a_x = -2\omega \cos \varphi (v_0 \cos \alpha - gt)$$

$$dv_x = -2\omega \cos \varphi (v_0 \cos \alpha - gt) dt$$

$$C + v_x = -2\omega \cos \varphi \left(v_0 \cos \alpha t - \frac{gt^2}{2} \right), \quad C = -v_0 \sin \alpha$$

$$\int_0^{\frac{2v_0 \cos \alpha}{g}} dv_x = \int_0^{\frac{2v_0 \cos \alpha}{g}} \left(v_0 \sin \alpha - 2\omega \cos \varphi \left(v_0 \cos \alpha t - \frac{gt^2}{2} \right) \right) dt$$

$$0 = v_0 \sin \alpha t - 2\omega \cos \varphi \left(v_0 \cos \alpha \frac{t^2}{2} - \frac{gt^3}{6} \right) \Big|_0^{\frac{2v_0 \cos \alpha}{g}}$$

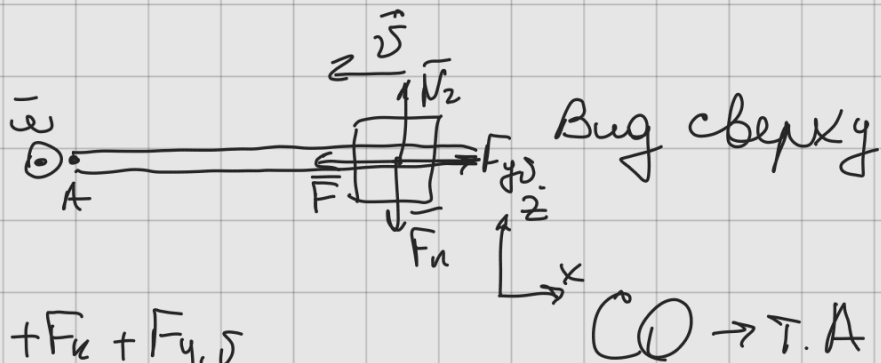
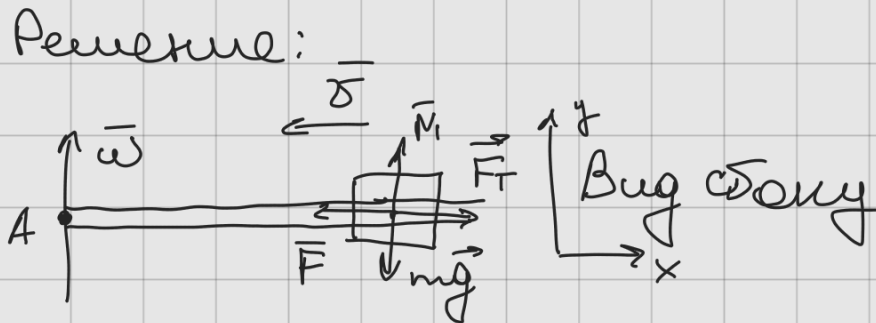
$$v_0 \sin \alpha = \omega \cos \varphi v_0 \cos \alpha t - \frac{\omega \cos \varphi}{3} gt^2$$

$$v_0 \sin \alpha = \omega \cos \varphi \cdot v_0 \cdot \frac{\cos \alpha \cdot 2v_0 \cos \alpha}{g} - \frac{\omega \cos \varphi}{3} \cdot \frac{4v_0^2 \cos^2 \alpha}{g}$$

$$\sin \alpha = \frac{2}{3g} \omega \cos \varphi v_0 \cos^2 \alpha \approx 2,47 \cdot 10^{-4} \text{ рад}$$

N&6

Дано:
 m, ω, v, μ
 Найти: A



$$F_x = m a_{x.c.o} + F_x + F_y v$$

$\parallel$ $\parallel$ $\nwarrow m\omega^2 R$
 0 $2m[\vec{v} \times \vec{\omega}]$

$$2) \text{ } 0y: N_1 = mg \quad 0z: N_2 = F_n = 2m\omega v$$

$$0x: F_T + F_y v = F$$

$$F_T = \mu N = \mu \sqrt{N_1^2 + N_2^2} = \mu m \sqrt{4\omega^2 v^2 + g^2}$$

$$A = \int F dx = \int_0^L (\mu m \sqrt{4\omega^2 v^2 + g^2} + m\omega^2 x) dx$$

$$A = \mu m L \sqrt{4\omega^2 v^2 + g^2} + \frac{m\omega^2 L^2}{2}$$

N&7

Дано:

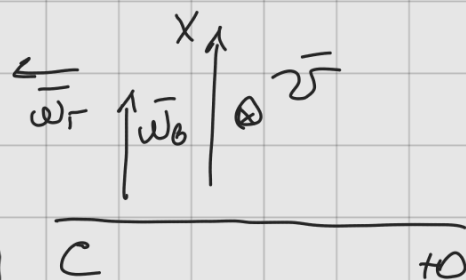
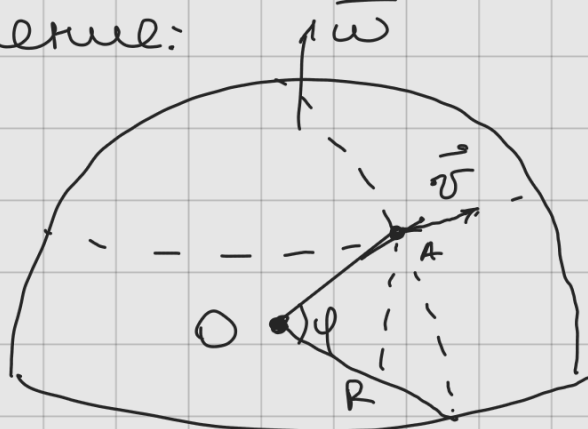
$$\varphi = 60^\circ$$

$$v = 10 \text{ м/с}$$

P_0

Найти: P

Решение:



1) Показываем тело: $P_0 = mg - m\omega^2 R \cos^2 \varphi$

2) $F_n = -m A_{\text{центр}} + F_n + F_{y.б}$ ($CO \rightarrow T.A.$)

$$-m(-\omega^2 R \cos^2 \varphi - \frac{v^2}{R})$$

$$\omega^2 R \cos^2 \varphi < g$$

$$P = mg - m\omega^2 R \cos^2 \varphi - \frac{mv^2}{R} - 2m v \omega \cos \varphi$$

$$\approx mg - \frac{mv^2}{R} - 2m v \omega \cos \varphi = P_0 \left(1 - \frac{v^2}{gR} - \frac{2v\omega \cos \varphi}{g} \right)$$

N&P

Дано:

$$v_0 = 900 \text{ м/с}$$

$$L = 18 \text{ км}$$

$$\varphi = 60^\circ$$

Найти: Δl

Решение:

1) $t = \frac{L}{v_0}$

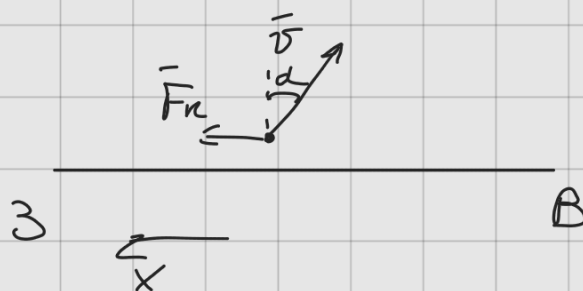
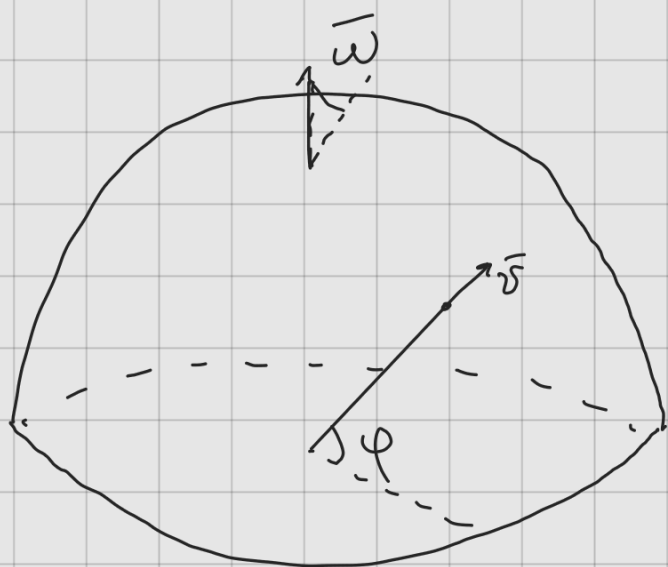


$$F_n = 2m v \omega \sin \varphi$$

$$dv_x = 2 v \omega \sin \varphi$$

$$S_x = v \omega \sin \varphi t^2 = \frac{v \omega \sin \varphi L^2}{v_0^2} = \frac{\omega \sin \varphi L^2}{v_0}$$

$$= 22,6 \text{ (м)}$$



$$v_y = v \cos \alpha - g t \rightarrow t_{\pi} = \frac{2v \cos \alpha}{g}$$

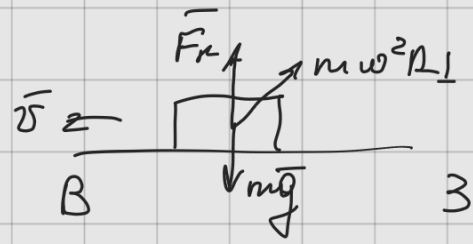
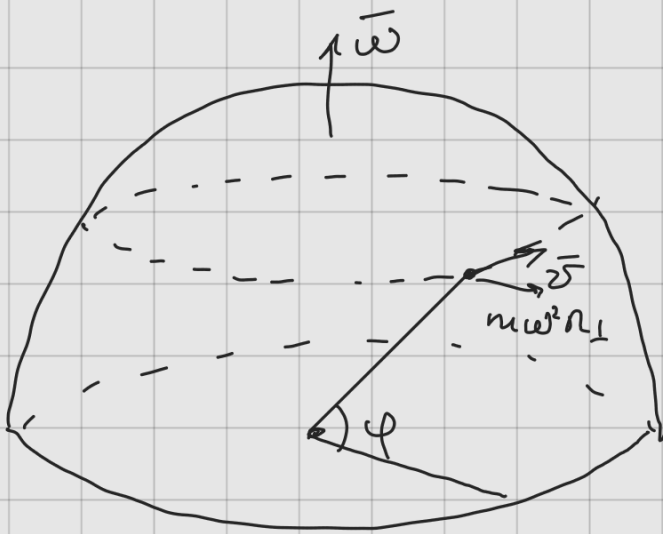
$$a_x = 2m(v \cos \alpha - g t) \cdot \omega \cdot \cos \varphi$$

$$v_x = 2m(v \cos \alpha t - \frac{g t^2}{2}) \omega \cos \varphi - v \sin \alpha$$

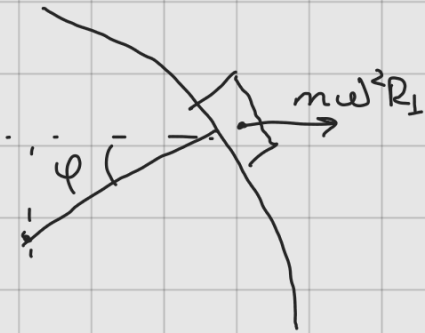
$$S_x = m(v \cos \alpha t^2 - \frac{g t^3}{3}) \omega \cos \varphi - v \sin \alpha t = 0$$

$$v \sin \alpha = (m v \cos \alpha - \frac{2v \cos \alpha}{g} - \frac{g m \cdot 4v^2 \cos^2 \alpha}{3g^2}) \omega \cos \varphi$$

$$\sin \alpha = \frac{2v m \cos^2 \alpha}{g} \omega \cos \varphi - \frac{m v \cos \varphi}{3g} \cdot 4 \cos^2 \alpha$$



$$F_K = 2m v \omega \cos \varphi$$



$$1) P_0 = mg - m\omega^2 R \cos^2 \varphi \approx mg$$

$$2) A = mg - 2m\omega \cos \varphi v - m\omega^2 R \cos^2 \varphi - \frac{mv^2}{R} =$$